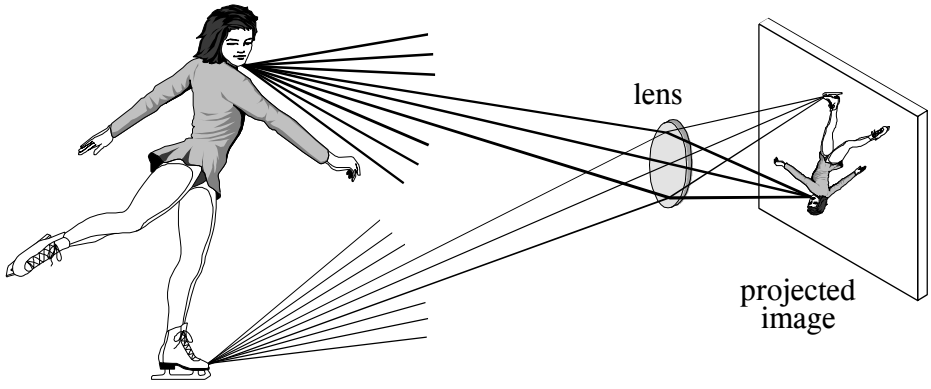

FIGURE 23-2
Focusing by a lens. A lens gathers light expanding from a point source, and force it to return to a point at another location. This allows a lens to project an image onto a surface.

Refraction in the lens changes the direction of the individual photons, depending on the location and angle they strike the glass/air interface. These direction changes cause light expanding from a single point to return to a single point on the projection screen. All of the photons that reflect from the toe *and* pass through the lens are brought back together at the "toe" in the projected image. In a similar way, a portion of the light coming from *any* point on the object will pass through the lens, and be focused to a corresponding point in the projected image.

Figures 23-3 and 23-4 illustrate the major structures in an electronic camera and the human eye, respectively. Both are light tight enclosures with a lens mounted at one end and an image sensor at the other. The camera is filled with air, while the eye is filled with a transparent liquid. Each lens system has two adjustable parameters: **focus** and **iris diameter**.

If the lens is not properly focused, each point on the object will project to a circular region on the imaging sensor, causing the image to be blurry. In the camera, focusing is achieved by physically moving the lens toward or away from the imaging sensor. In comparison, the eye contains two lenses, a bulge on the front of the eyeball called the cornea, and an adjustable lens inside the eye. The cornea does most of the light refraction, but is fixed in shape and location. Adjustment to the focusing is accomplished by the inner lens, a flexible structure that can be deformed by the action of the *ciliary muscles*. As these muscles contract, the lens flattens to bring the object into a sharp focus.

In both systems, the *iris* is used to control how much of the lens is exposed to light, and therefore the brightness of the image projected onto the imaging sensor. The iris of the eye is formed from opaque muscle tissue that can be contracted to make the *pupil* (the light opening) larger. The iris in a camera is a mechanical assembly that performs the same function.